FASHN VTON training workflow

scripts/train.py and scripts/train_from_scratch.py

1 CSV manifest

person_image, garment_image, target_image,
category, optional garment_photo_type.
Paths are relative to the CSV directory.

2 Frozen preprocess (not trained)

DWPose on the person. Flat-lay garments
get dummy keypoints; model photos get
a real pose. The parser builds the
clothing-agnostic person and garment.
Resize-pad to 864 x 576, scale to [-1, 1].

3 TryOnModel (the only trained net)

Fine-tune loads model.safetensors.
From scratch calls init_tryon_model.
MMDiT: hidden 1280, 10 heads, 8 double
blocks, 16 single blocks, 4 patch-mixer
blocks, patch size 12.

4 Rectified-flow step

x_t = (1 - t) * noise + t * target.
Target velocity is target - noise.
Loss is mean squared error. About 10%
of steps drop the condition (CFG).
AdamW, clip grads at 1.0, warmup-cosine.

5 Write the output directory

train_state_*.pt can resume the run.
Final weights: model_finetune_last
or model_scratch_last .safetensors.

What runs where

CUDA GPU, when PyTorch can see one:
TryOnModel, DWPose, human parser.
bfloat16 on Ampere and newer.
CPU: open images and resize them.
--num-workers defaults to 0.
No CUDA: same loop on CPU, float32.
That CPU path is a smoke test.

How long

No timer in the script. Default is
1 epoch, batch size 1, so steps equal
the CSV row count, unless --epochs
or --max-steps is set.
VITON-HD test is 2,032 rows, so one
default epoch is 2,032 steps.
Seconds per step are not measured.
Each step is one forward-backward
of the MMDiT, plus pose and parsing.

Defaults

Fine-tune: lr 1e-5, warmup 500,
checkpoint every 500 steps.
From scratch: lr 1e-4, warmup 2000,
checkpoint every 1000, EMA 0.9999.
Install: Ubuntu x86_64, Python 3.12,
PyTorch 2.11.0+cu128. The NVIDIA
driver must support CUDA 12.8.

Step detail

Start

cd fashn-vton-1.5, then python scripts/train.py with --manifest, --aux-weights-dir, and --output-dir. train_from_scratch.py is the same entry with --from-scratch forced on. Aux weights are dwpose/ plus, for a fine-tune, model.safetensors.

Each row

The dataset loads three photos and a category (tops, bottoms, or one-pieces). garment_photo_type is model or flat-lay. Preprocessing matches TryOnPipeline: pose images, a clothing-agnostic person, and a garment image. segmentation-free is on by default, so the person garment region is not masked out.

The update

A uniform timestep in (0, 1] is shifted with mu 1.5, the same bias inference uses. The model sees the noisy target plus the frozen condition tensors. Classifier-free dropout uses cond_mask = random > 0.10. Gradients accumulate for --grad-accum steps (default 1) before AdamW steps.

Hardware the code will use

It does not pick a card by name. torch.cuda.is_available() selects the GPU; otherwise CPU. On CUDA, bfloat16 is used when torch.cuda.is_bf16_supported() is true. ONNX Runtime GPU runs DWPose. FashnHumanParser uses the same CUDA device. The prepared wheels are manylinux x86_64 for CUDA 12.8, so the machine is an Ubuntu box with a matching NVIDIA driver. This Windows tree is only where the bundle was packed.

Duration

Wall-clock time is steps times the time of one step, and that per-step time is not stored in the repo. Count steps as rows × epochs / batch size, capped by --max-steps. Fine-tune checkpoints land every 500 steps; from-scratch checkpoints every 1000. A 2,032-row test manifest at the defaults is 2,032 GPU steps, not a fixed number of hours.